

Session 6 MM-MBC

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21-09-2026

Membrane Structure

- Membrane composition
- Membrane proteins
- Summary

Nuclear-Cytoplasmic import/export

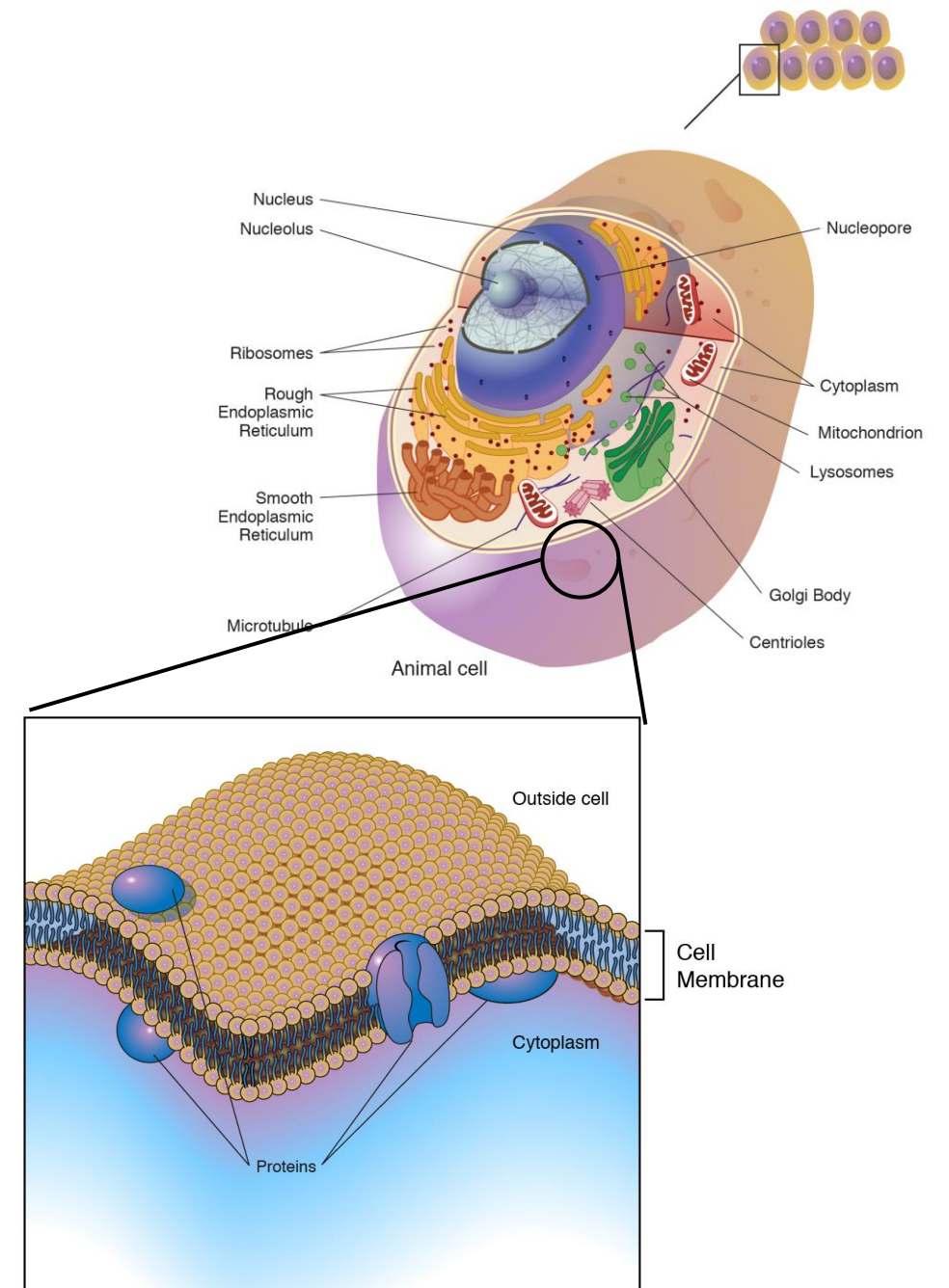
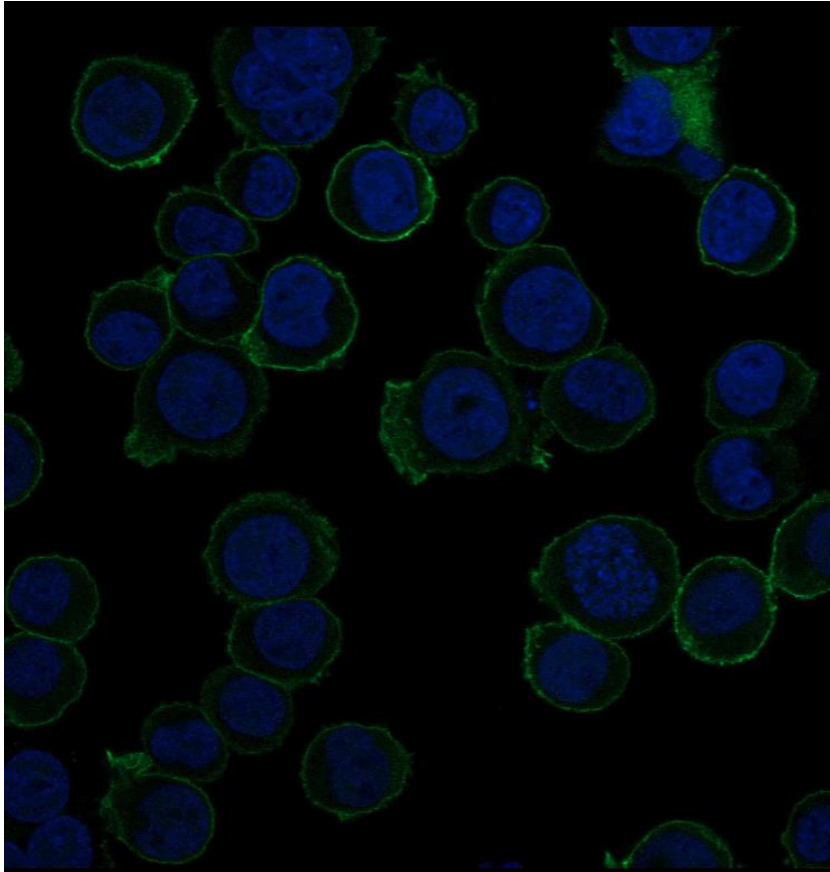
- Nuclear Pore Complexes
- Nuclear transport of ions, mRNAs and Proteins
- Diseases associated to nuclear transport defects
- Techniques
- Summary

Membrane Structure

Chapter 10

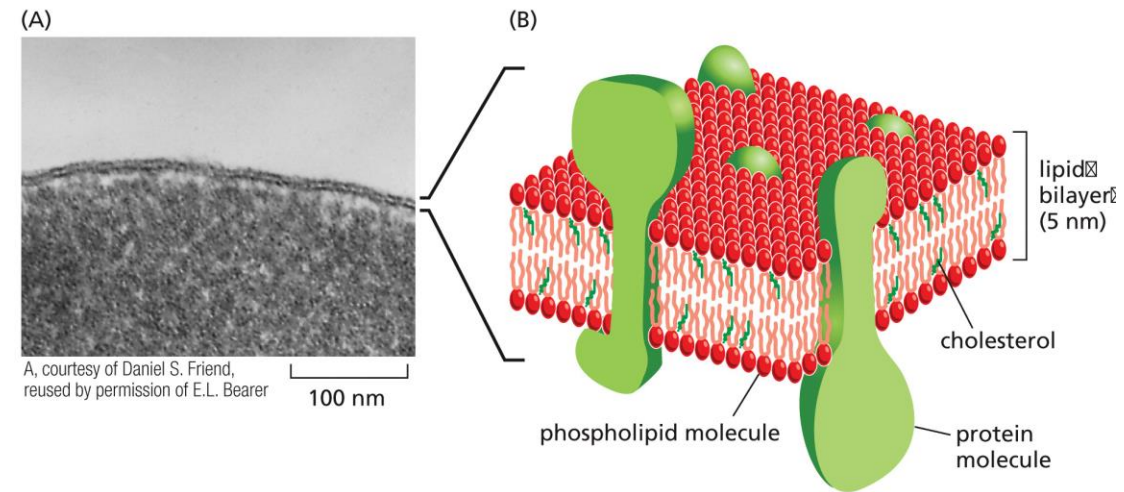
The Plasma Membrane

Plasma Membrane
Nucleus

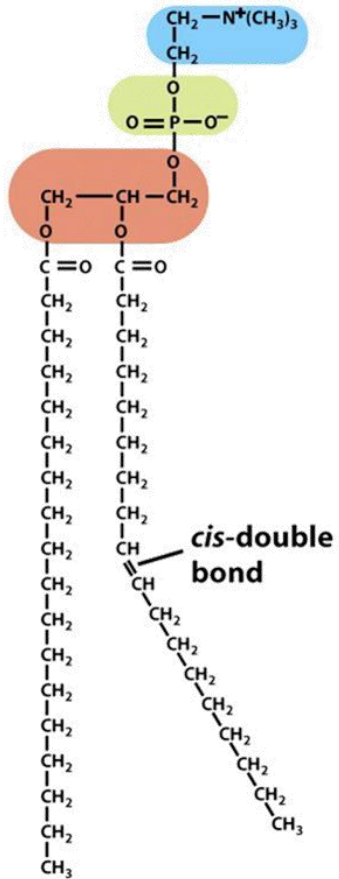


The Plasma Membrane

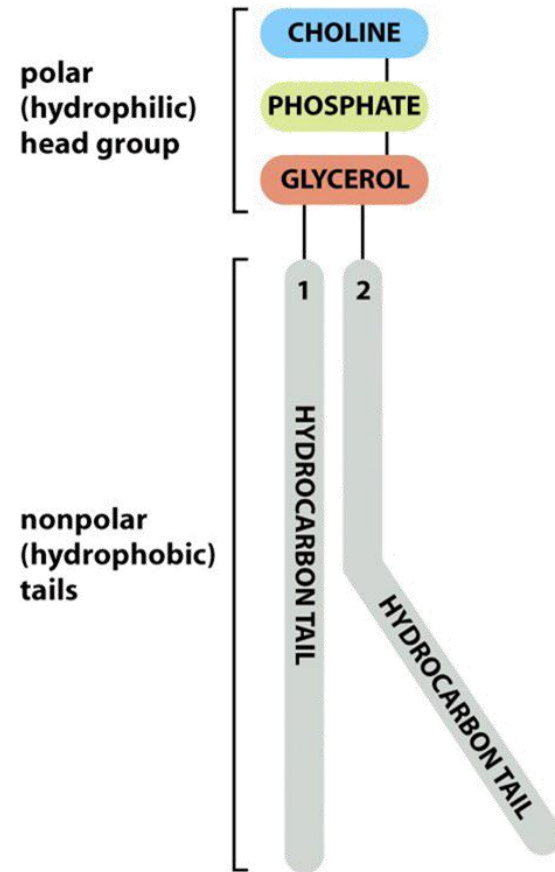
- Separates and protects the living cell from its surrounding
- Exhibits selective permeability, allowing some substances to cross it more easily than others
- Consists of a lipid bilayer with proteins embedded in it
- Fluid
- Involved in several cellular processes:
 - Cell-cell and cell-substrate adhesion
 - Ion-conductivity
 - Cell-signaling
 - Internalization and externalization of components (Endocytosis and Exocytosis)



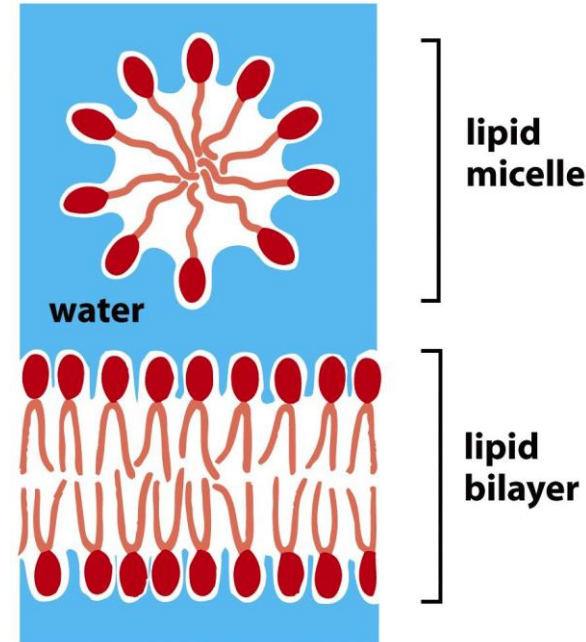
Lipid bilayer



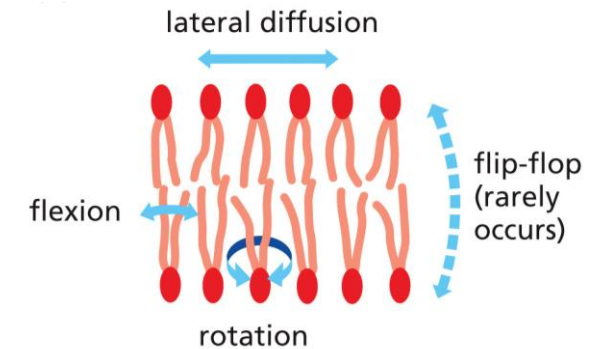
Phospholipids are the most abundant lipid in the plasma membrane...



...and are amphipathic molecules containing hydrophobic and hydrophylic regions



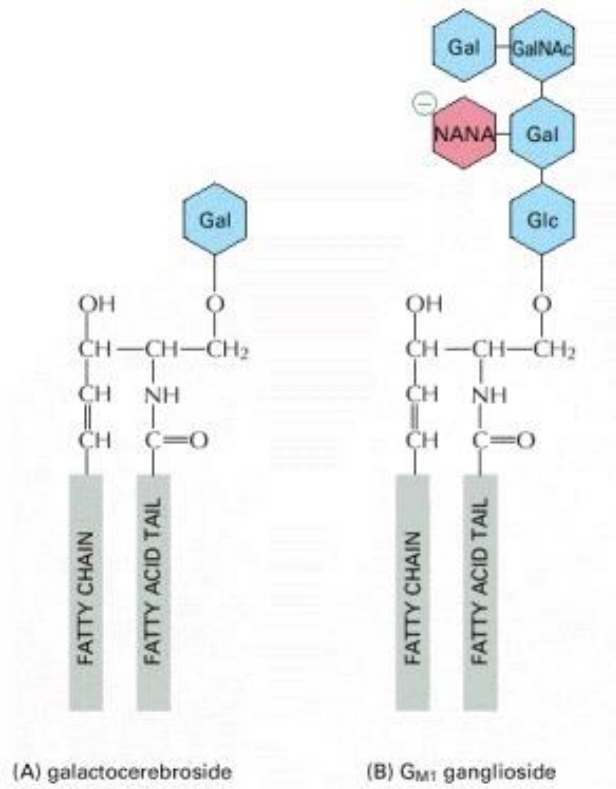
Phospholipids spontaneously form bilayers in aqueous environments with the hydrophilic polar head to the external



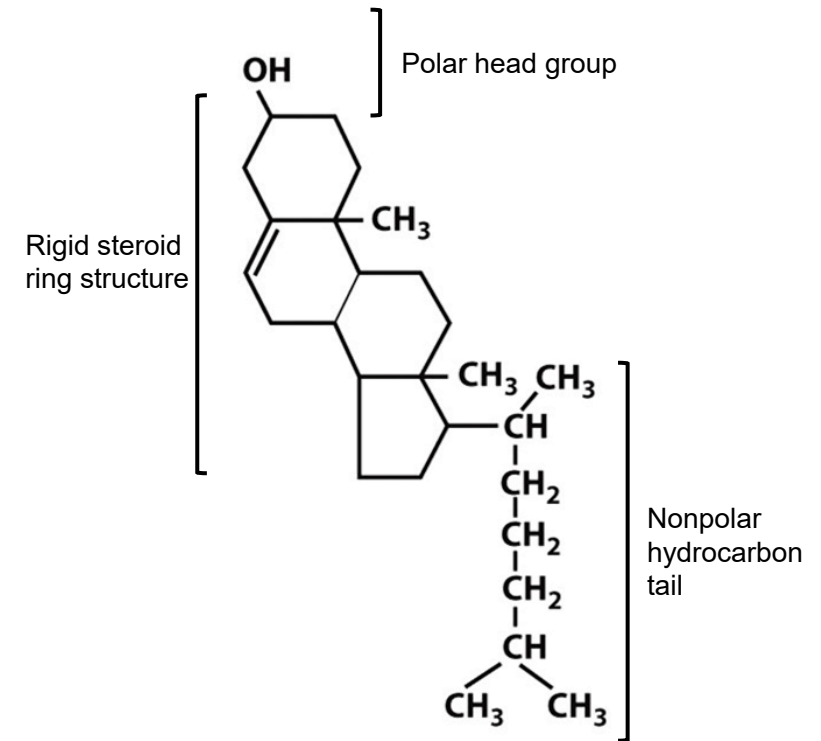
Lipid bilayer is a 2D fluid structure

Lipid bilayer

Lipid bilayer consists also of glycolipids and sterols



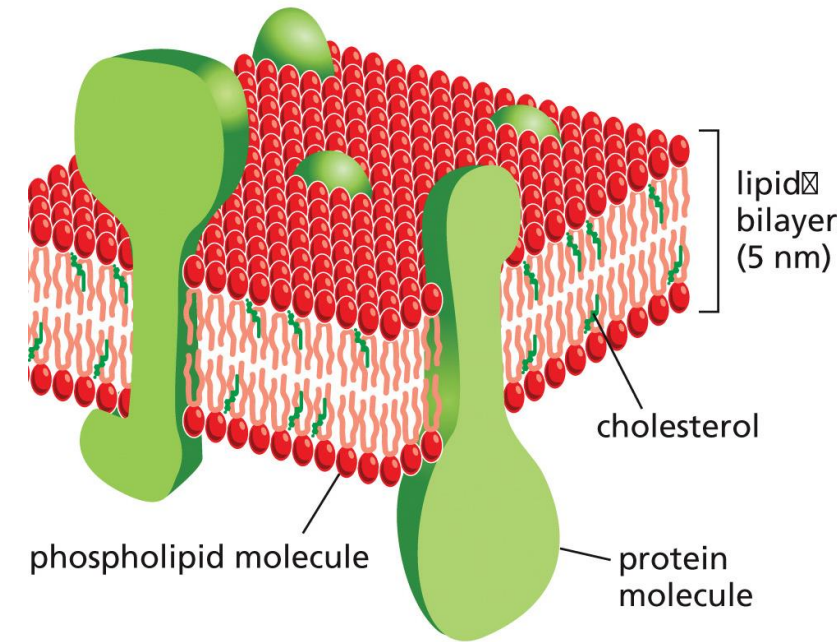
Sugars attached instead of phosphate linked group



Cholesterol

Fluid mosaic model of membranes

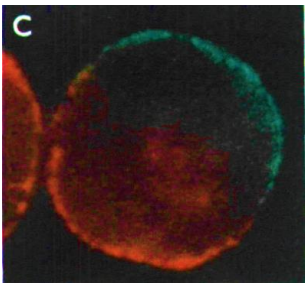
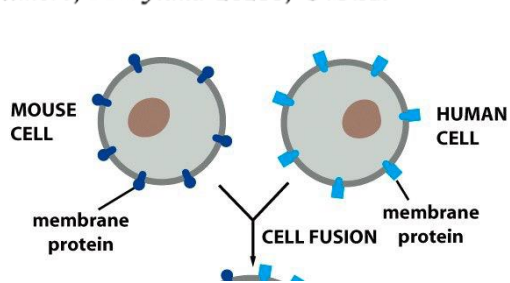
- The fluid mosaic model states that a membrane is a fluid structure with a mosaic of various proteins embedded in it.
- The fluidity of a lipid bilayer depends on its composition
 - Different cell types or cell organelles have different lipid composition
- Lipid Rafts (LR) are clusters of cholesterol and glycosphingolipids present in the membrane
 - LRs can move freely within the membrane
 - Proteins embedded in LRs can be brought in proximity of other proteins
 - LRs can therefore regulate cell signaling



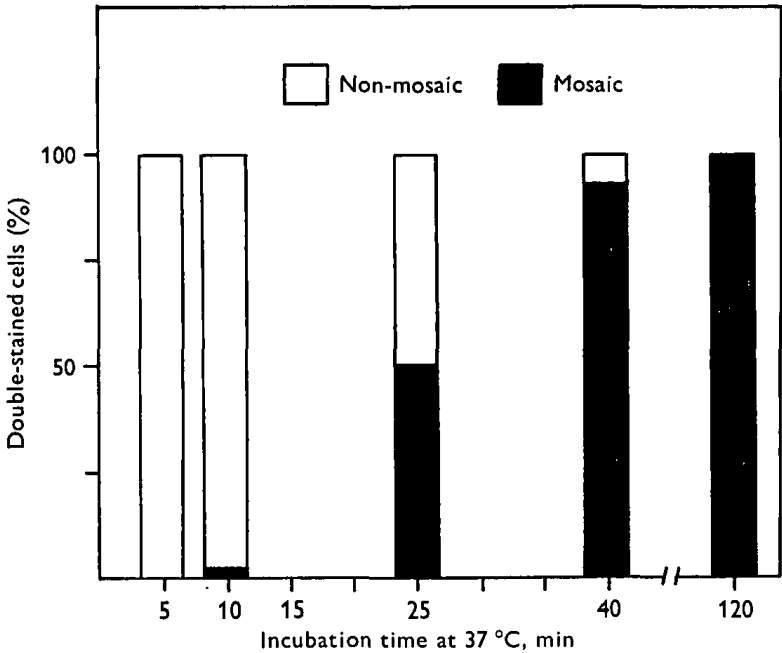
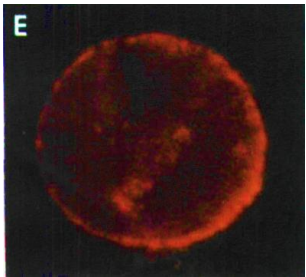
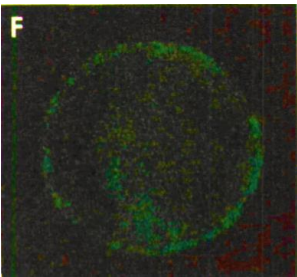
Fluid mosaic model of membranes: experimental proof

THE RAPID INTERMIXING OF CELL SURFACE ANTIGENS AFTER FORMATION OF MOUSE-HUMAN HETEROKARYONS

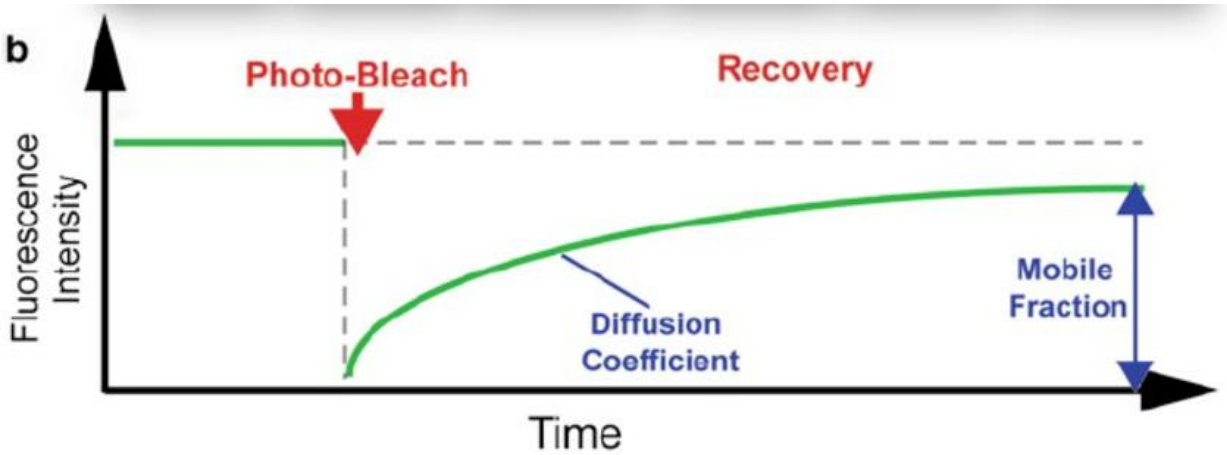
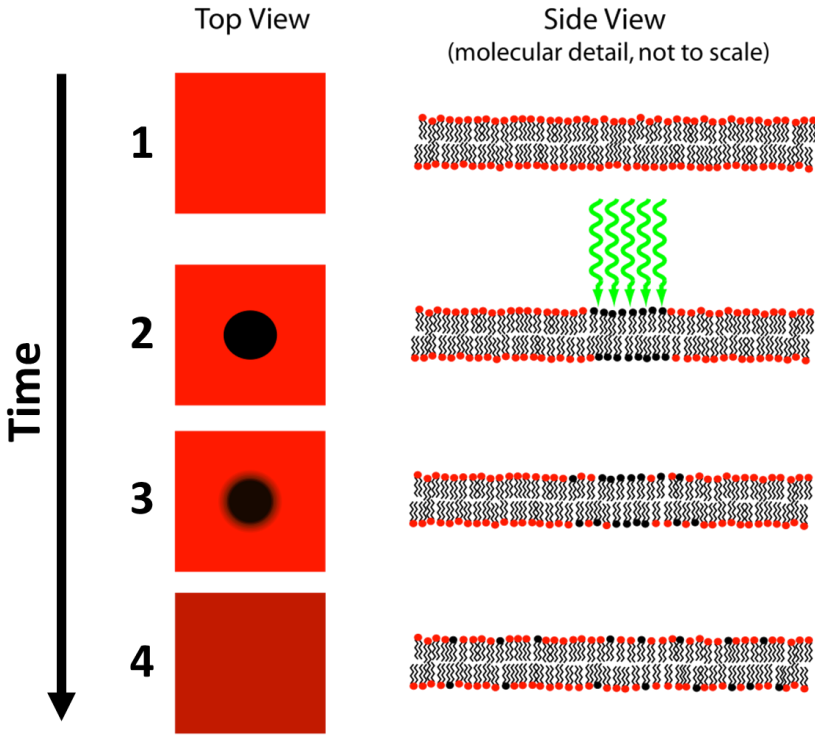
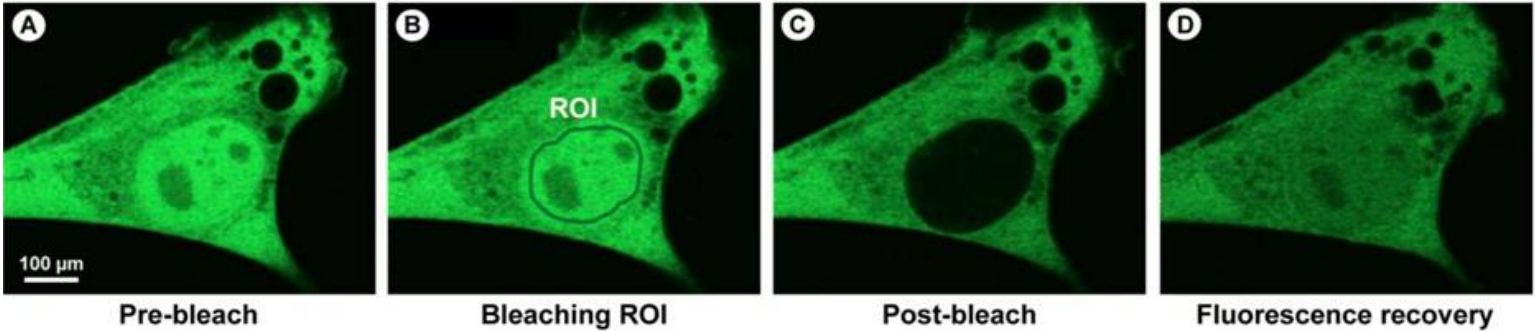
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Baltimore, Maryland 21218, U.S.A.*



heterokaryon

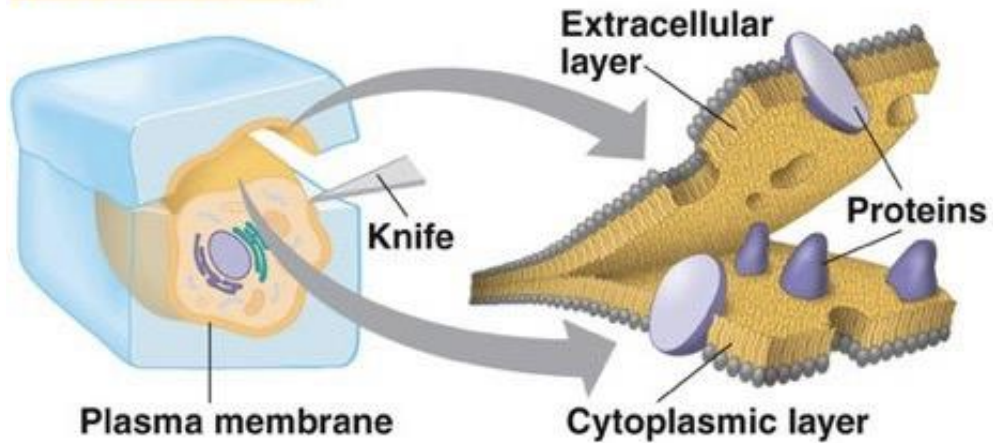


Fluorescence Recovery After Photo-Bleaching (FRAP)



Freeze-Fracture

TECHNIQUE



RESULTS



Inside of extracellular layer

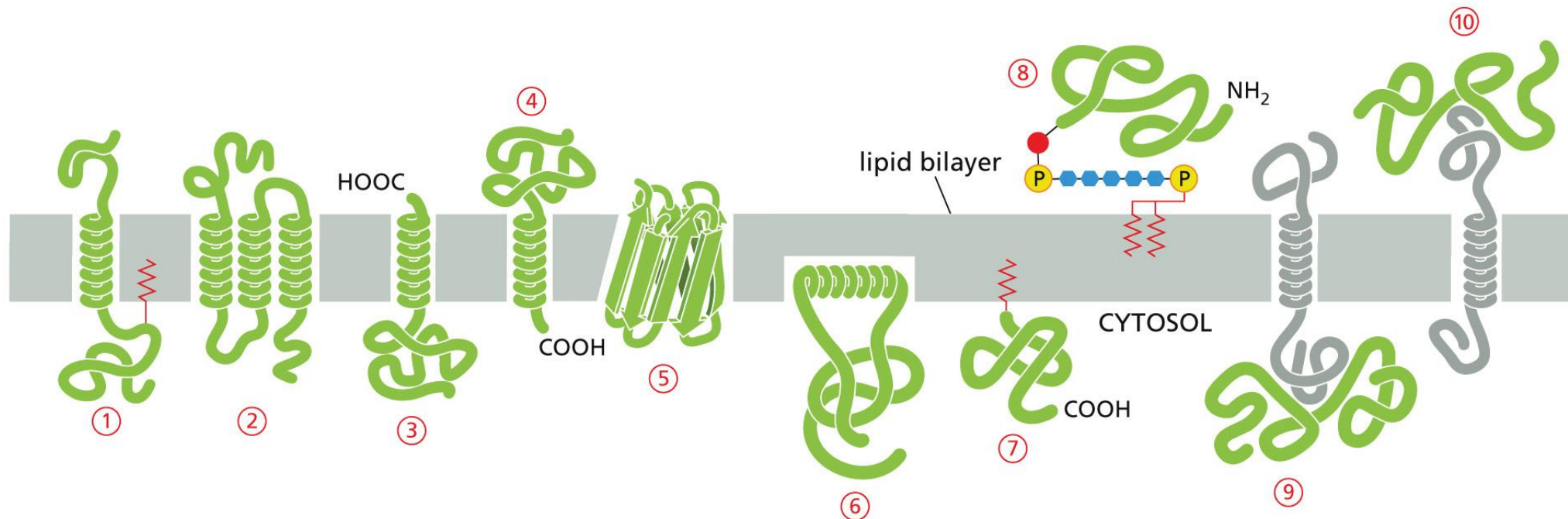


Inside of cytoplasmic layer

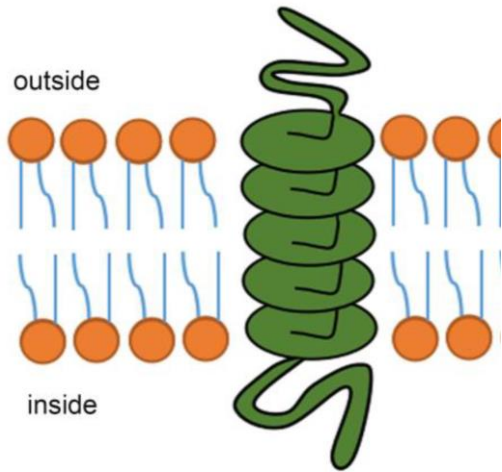
- Splits a membrane along the middle of the lipid bilayer.
- Membranes are then visualized under an electron microscopy.

Membrane Proteins and their functions

- Membrane Proteins
 - Transmembrane proteins (1, 2, 5)
 - Integral proteins (3,4)
 - Monotopic (6)
 - Lipid associated (7)
 - Glycosylphosphatidylinositol (GPI) anchor proteins (8)
- Membrane-Associated Proteins (9, 10)

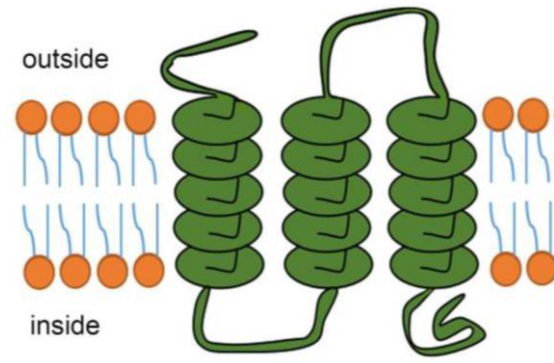


Transmembrane Proteins



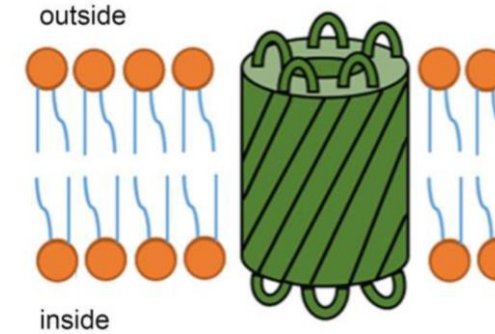
Single pass
(1 α -helix)

e.g.: Glycophorin A



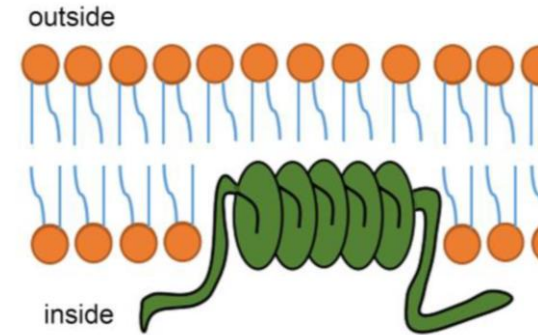
Multipass
(> 1 α -helix)

e.g.: Bacteriorhodopsin
(7 α -helices)



β -barrels

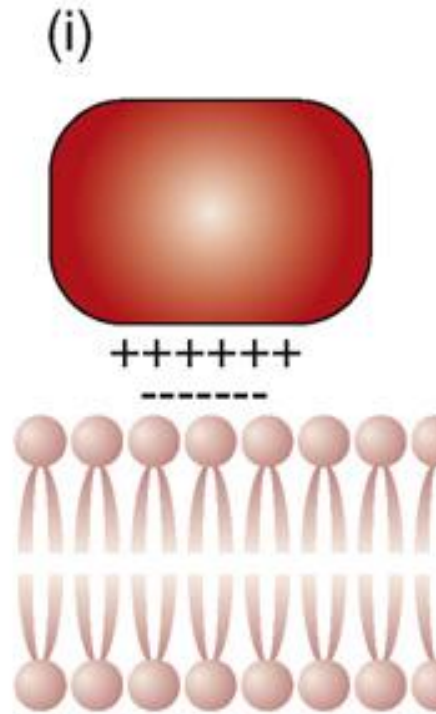
e.g.: Porins



Monotopic

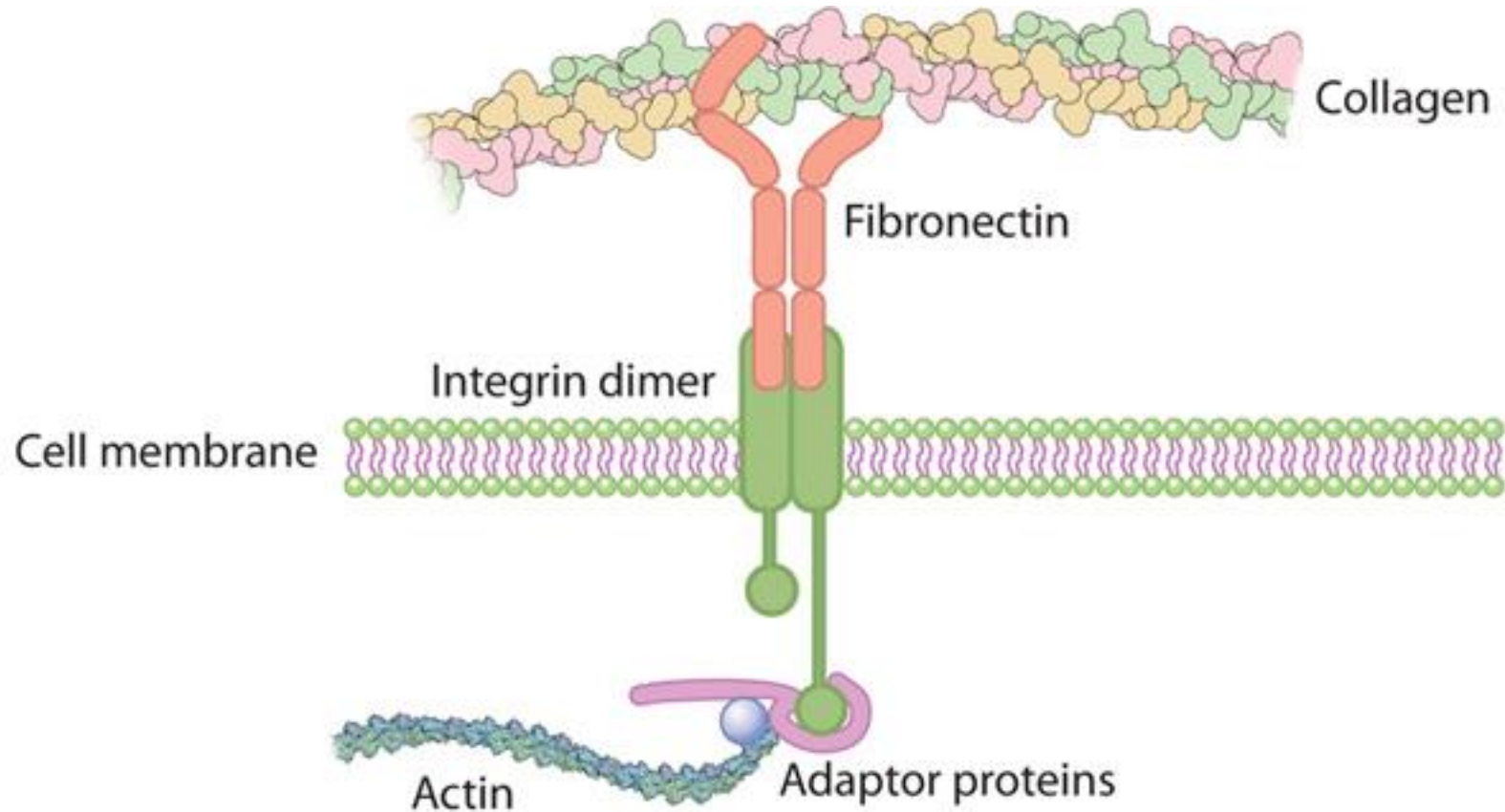
e.g.: COX-1
(Prostaglandin H2 synthase)

Lipid-associated and GPI proteins



- i. Electrostatic attraction of positively charged patches on protein surfaces (e.g.: Cytochrome c)
- ii. Covalent protein modifications (e.g.: GPI and Ras proteins)
- iii. Specific protein-lipid binding (e.g.: Annexins)
- iv. Non-specific hydrophobic associations (e.g.: MARCKS)

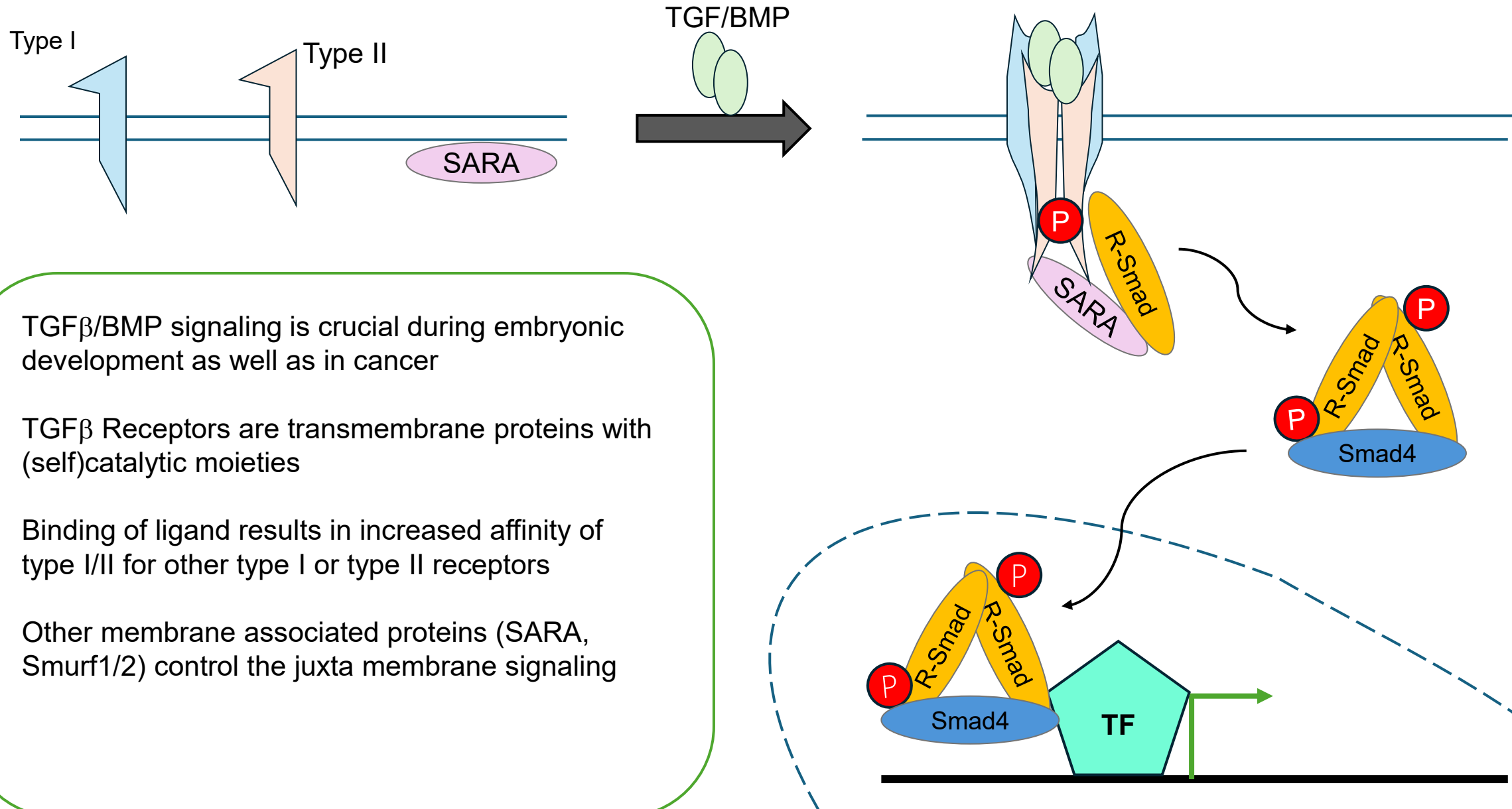
Membrane Associated Proteins



Functions of Membrane proteins

- Enzymatic activity
- Signal Transduction
- Cell-cell and Cell-substrate adhesion
- Intercellular communication
- Docking for cytoskeletal proteins

Membrane proteins and signaling: TGF β /BMP signaling



TGF β /BMP signaling is crucial during embryonic development as well as in cancer

TGF β Receptors are transmembrane proteins with (self)catalytic moieties

Binding of ligand results in increased affinity of type I/II for other type I or type II receptors

Other membrane associated proteins (SARA, Smurf1/2) control the juxta membrane signaling

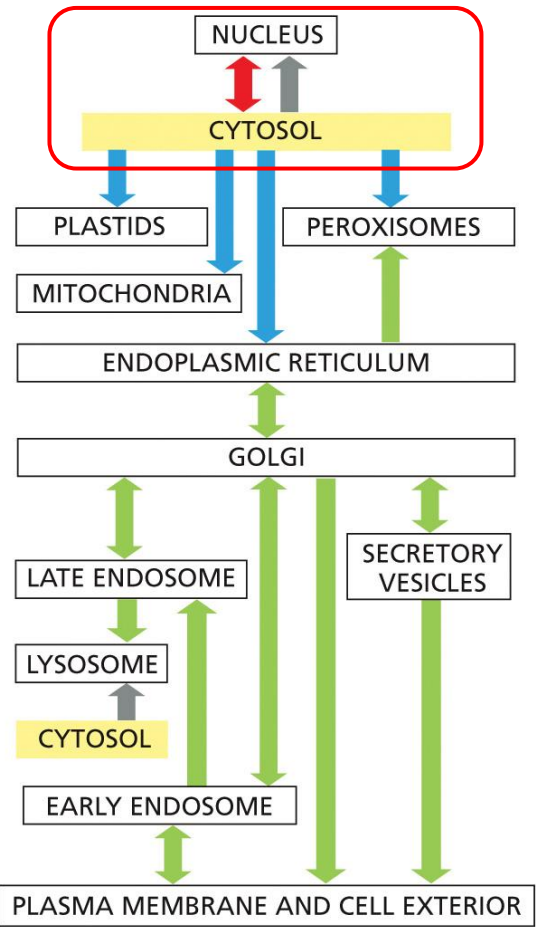
Summary:

- A cell is an open system exchanging molecules with the extracellular environment. This process is controlled by the plasma membrane.
- Plasma membranes are selectively permeable, regulating the enter/exit of specific molecules.
- Plasma membranes are composed of an amphipathic lipid bilayer where the hydrophobic tail is directed towards the inside and the hydrophilic head (negative) towards the outside.
- Membrane Proteins are embedded in the lipid bilayer.
- Plasma membranes are fluid and movement of lipid cluster (rafts) is allowed.

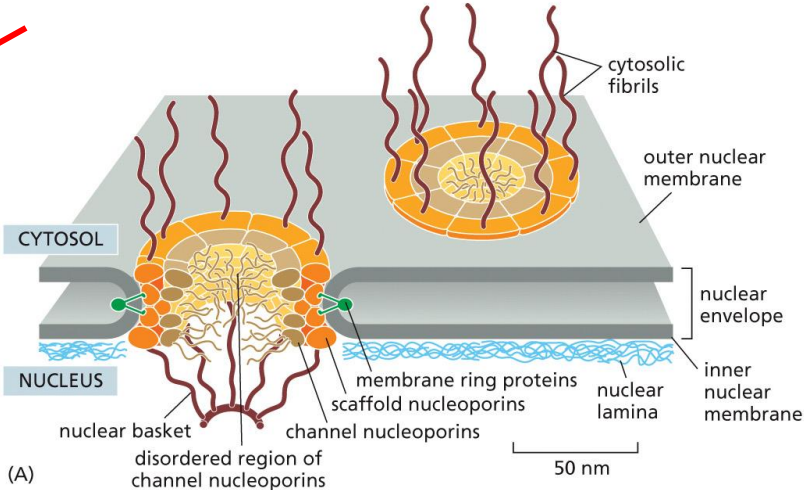
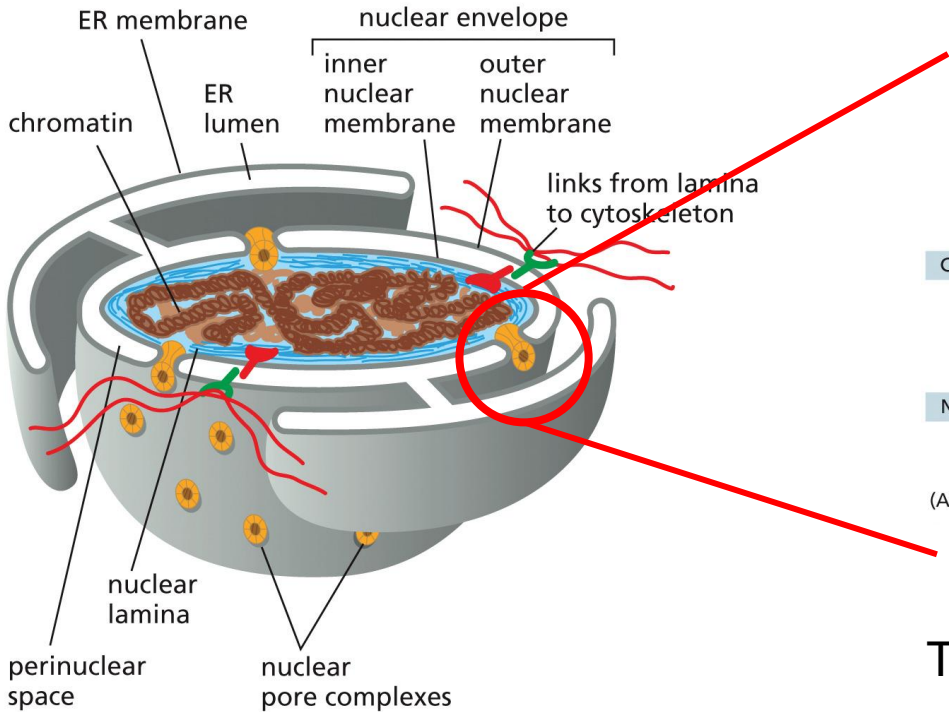
Nuclear-Cytoplasmic Import/Export

Chapter 12

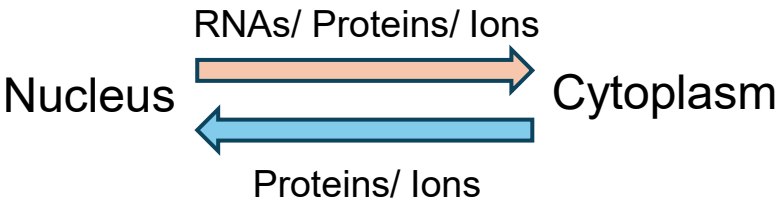
Nucleus-Cytoplasm import/export



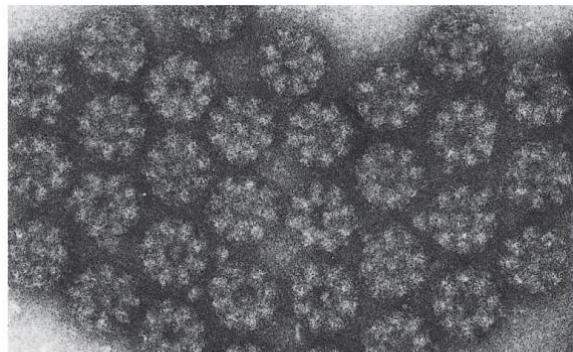
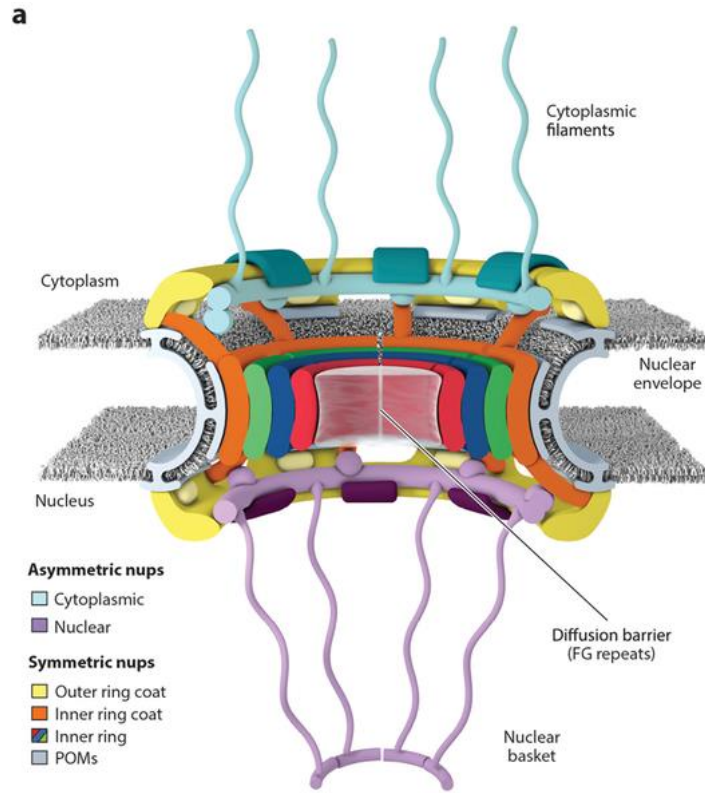
- █ = protein translocation
- █ = gated transport
- █ = vesicular transport
- █ = engulfment



The nuclear pores allows bidirectional nucleus/cytoplasm trafficking.



Nuclear Pore Complexes (NPC)

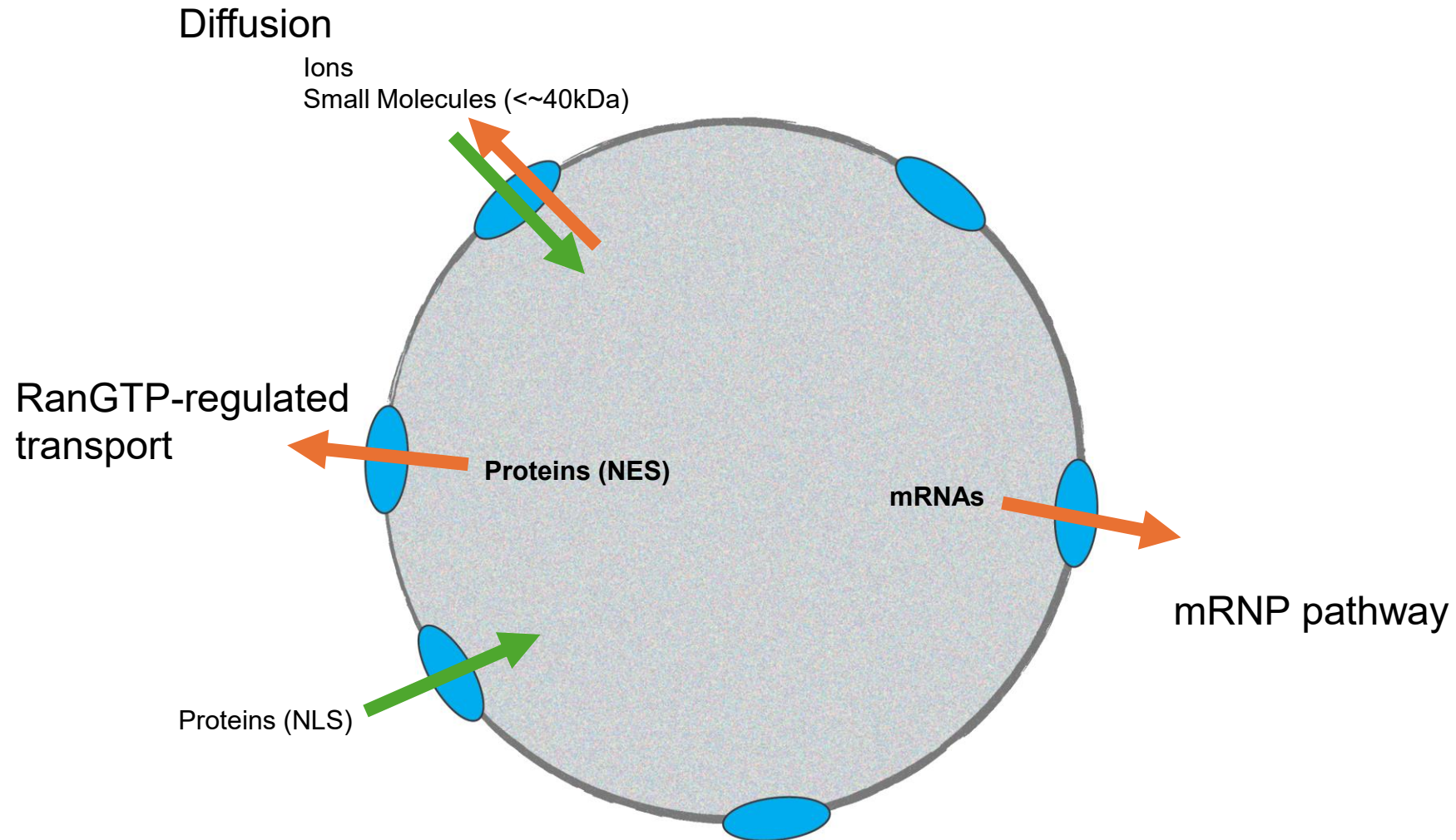


(D)

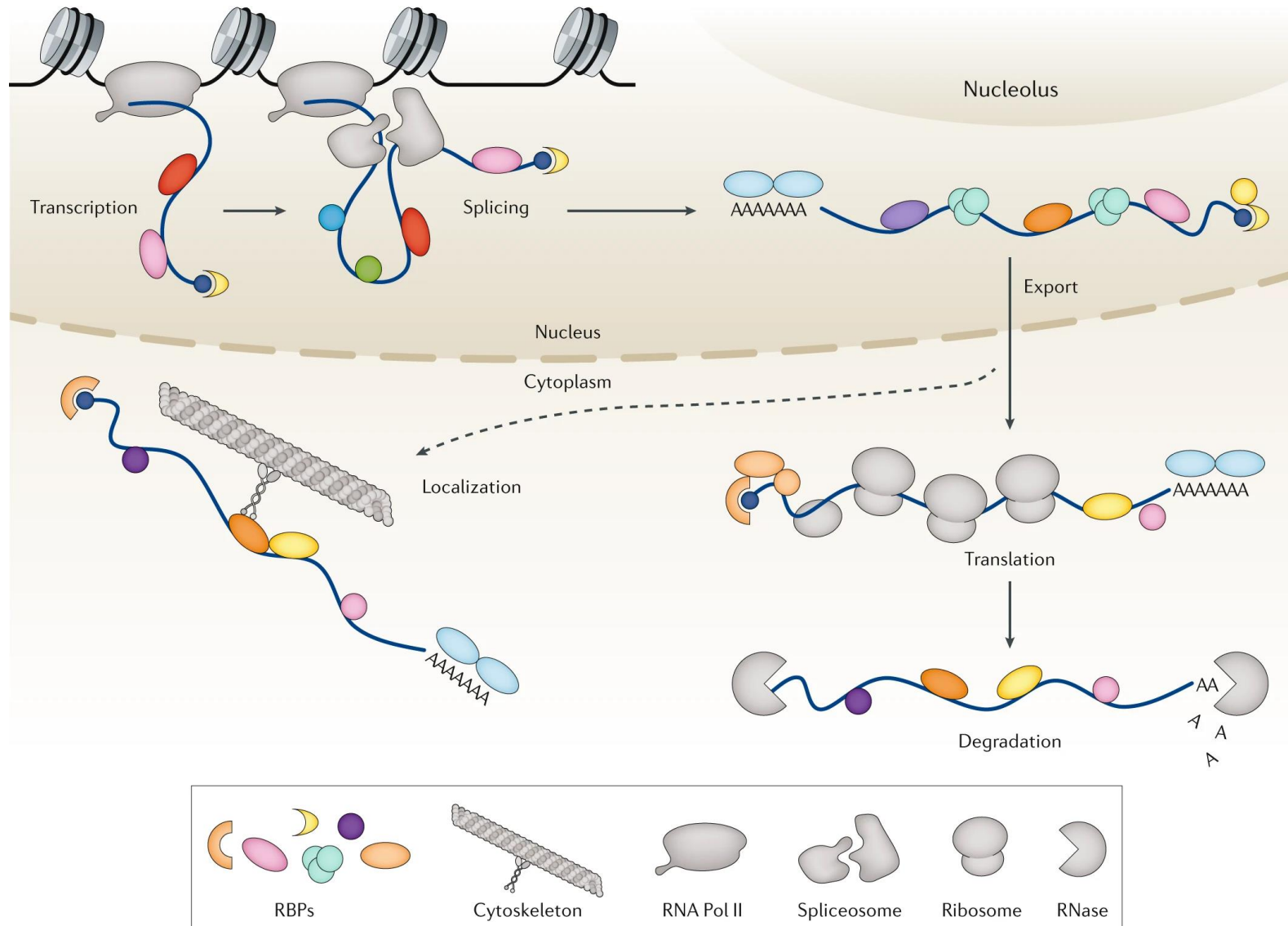
0.1 μm

- Large multiprotein structures embedded in the nuclear envelope
- Each NPC is composed of ~30/>60 (yeast/higher organisms) different nucleoporins organized in octagonal cylindrical symmetry
- NPCs have a cytoplasmic and a nuclear ring anchoring filaments that extend into the two compartments facilitating docking of transport factors and cargo molecules
- NPCs selectively allow the passage of molecules (RNAs, proteins, ribonucleoprotein complexes) >40kDa. Molecules up to ~40kDa can diffuse freely.
- The selective transport is mediated by FG repeats (phenylalanine-glycine) present in nucleoporins. The FG repeats interact with transport receptors to facilitate the passage of cargo molecules.
- Active transport through NPCs is energy-dependent and is mediated by Ran-GTPase which is responsible for the directionality of the transport.

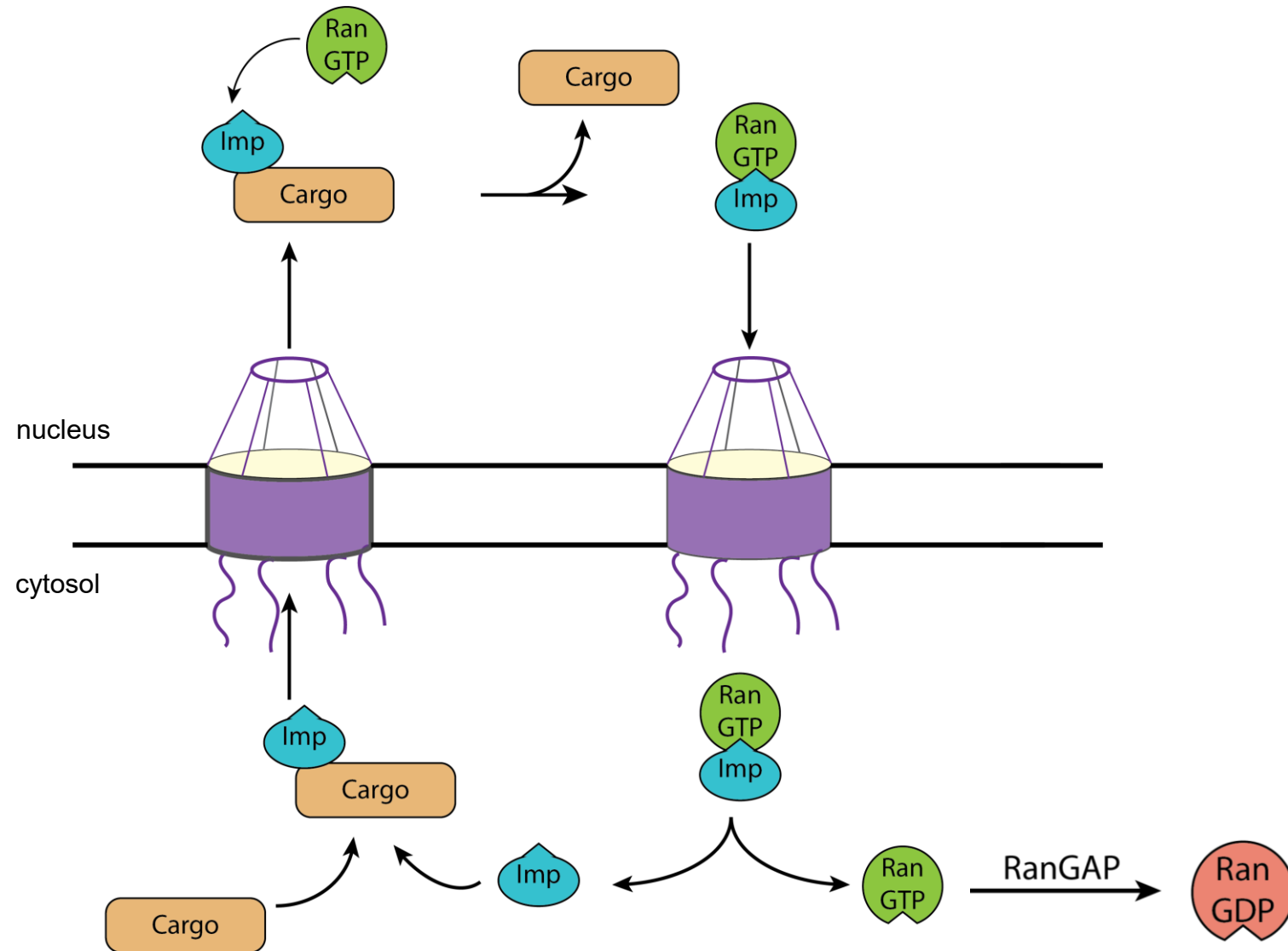
Nucleus-Cytoplasm import/export: a general overview



mRNP pathway

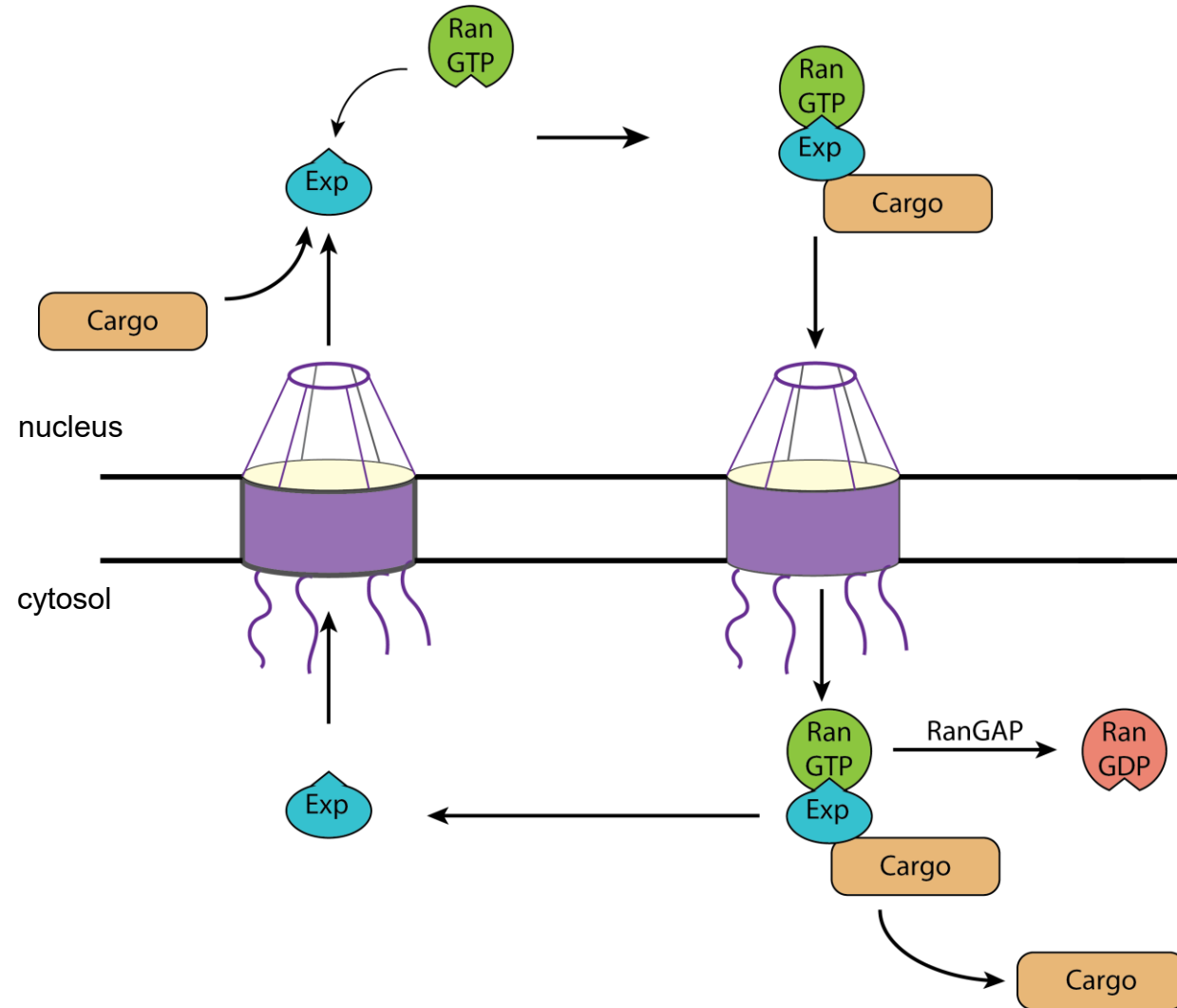


Ran-GTP regulated transport : nuclear import



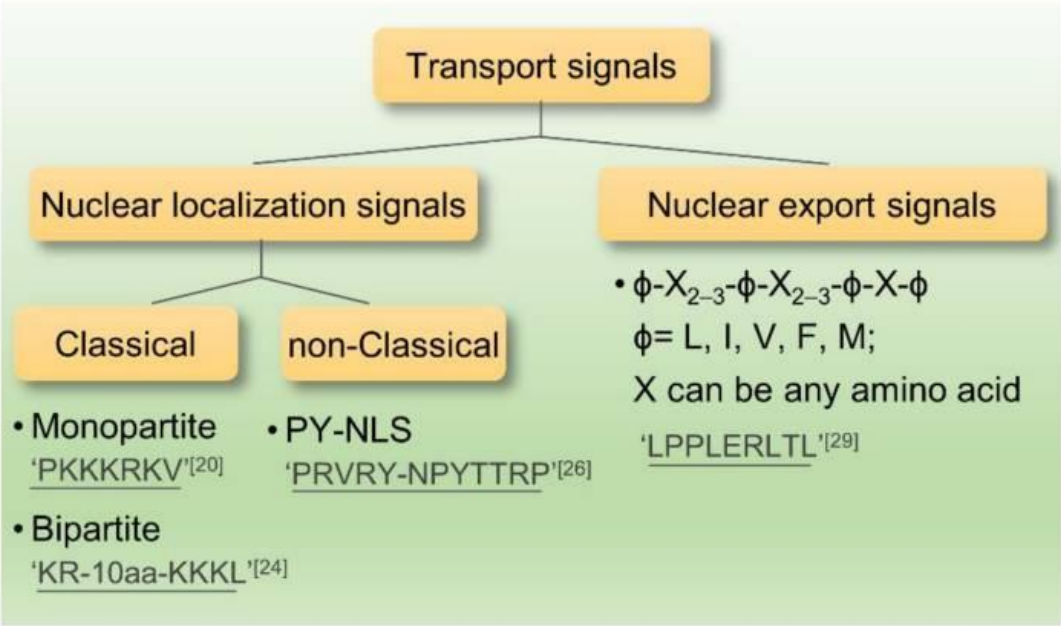
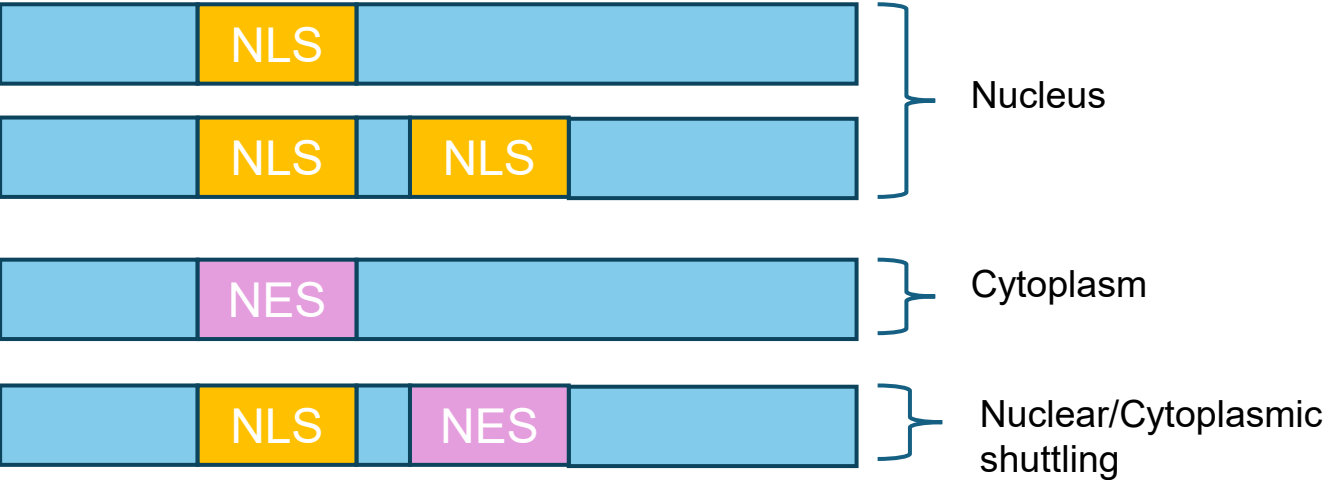
1. Cargo proteins containing a **Nuclear Localization Signal (NLS)** are recognized by **importins** (nuclear import receptors).
2. The importin-cargo complex binds to the nucleoporins of the NPC and is transported in the nucleus.
3. The RanGTP enzyme binds the importin and release the cargo in the nucleus.
4. The RanGTP-importin exits the nucleus.
5. RanGAP hydrolyzes Ran-GTP to Ran-GDP releasing the importin

Ran-GTP regulated transport: nuclear export

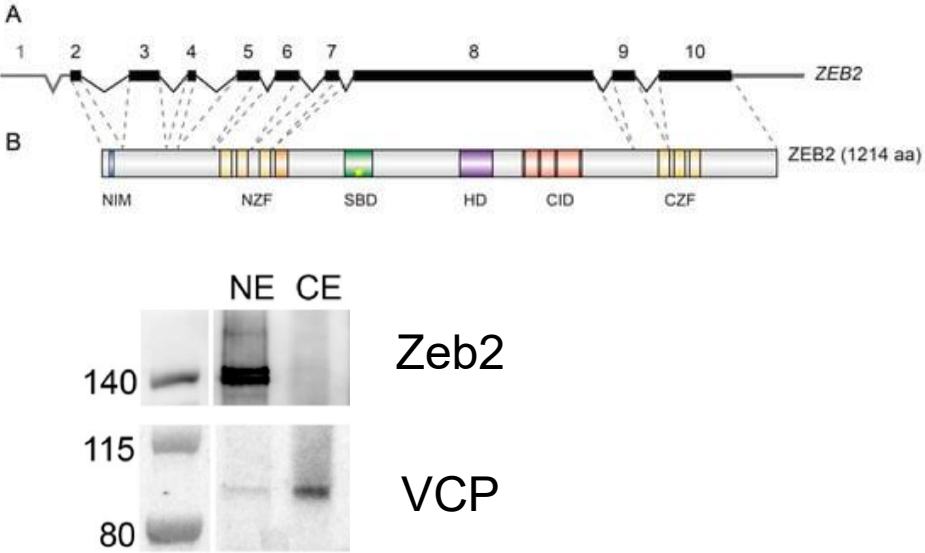


1. Proteins with a **Nuclear Export Signal (NES)** are recognized by **exportins** (nuclear export receptors), which require Ran-GTP for cargo binding.
2. The exportin-cargo-Ran-GTP complex is transported out of the nucleus through the NPC.
3. In the cytoplasm, RanGAP hydrolyzes Ran-GTP to Ran-GDP disassembling the export complex and releasing the cargo.
4. Exportins and Ran-GDP are then moved back to the nucleus.
5. In the nucleus, RanGEF converts Ran-GDP to Ran-GTP.

NLS & NES



Fu X, Liang C, Li F, Wang L, Wu X, Lu A, Xiao G, Zhang G.
Int J Mol Sci. 2018



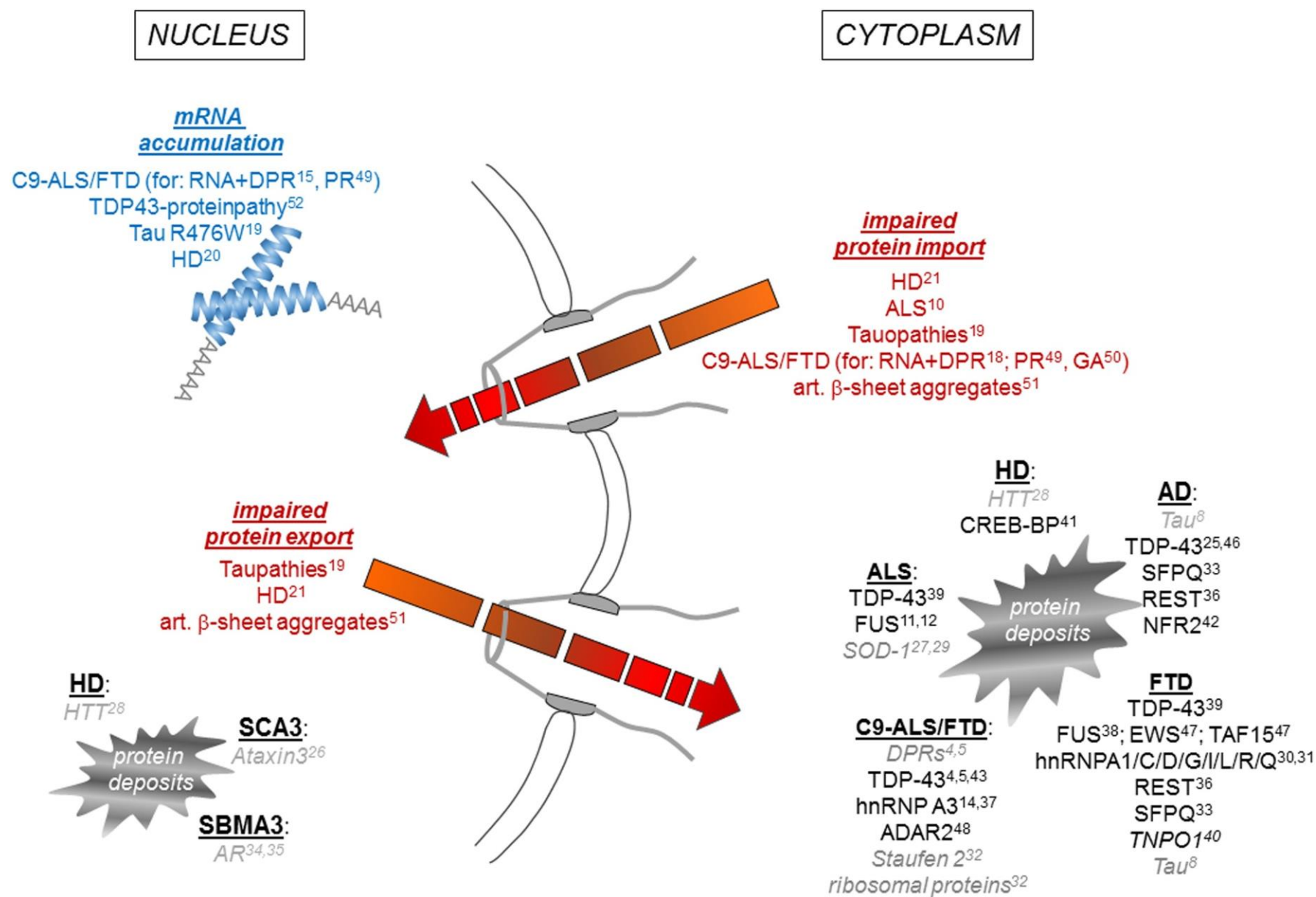
Feature	NLS	NES
Conserved domain cluster	Basic	Hydrophobic
Function	Import to the nucleus	Export from the nucleus
Receptor	Importins	Exportins
Activity	Active in the cytoplasm	Active in the nucleus

The TF Zeb2 is mainly localized in the nucleus

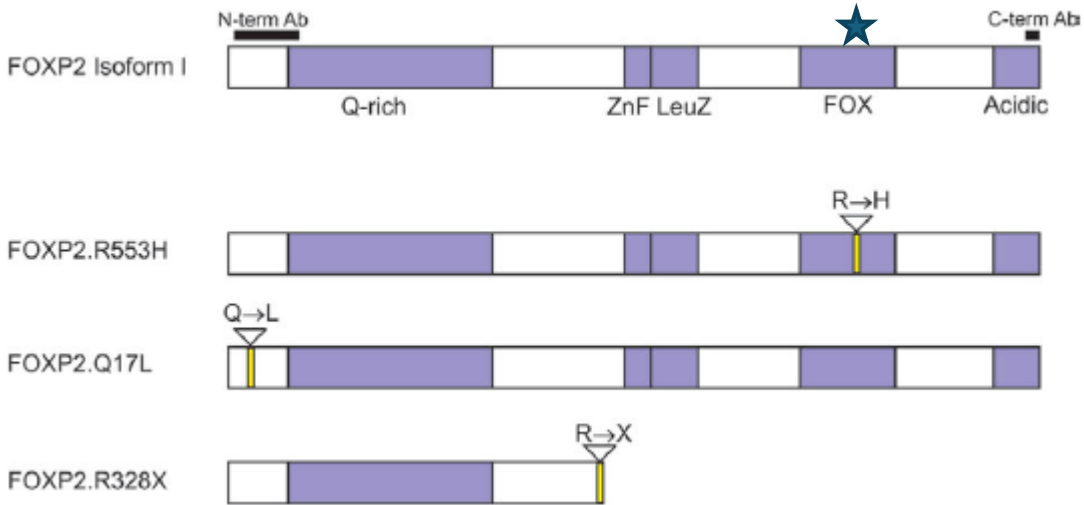
Nucleo-cytoplasmic transport: disease relevance (few examples)

Protein	Mechanisms	Consequences	Disease
SRY	Mutation of NLS	Loss of nuclear localization	Swyer Syndrome (gonadal dysgenesis)
FOXP2	Mutation of NLS	Loss of nuclear localization	Speech-language disorder (and ID)
FUS	Mutation of NLS	Cytoplasmic aggregates	ALS, Frontotemporal Dementia
FOXO	Mutation of NLS, PTMs	Cytoplasmic mislocalization	Cancer

Nucleo-cytoplasmic transport defects in NDD

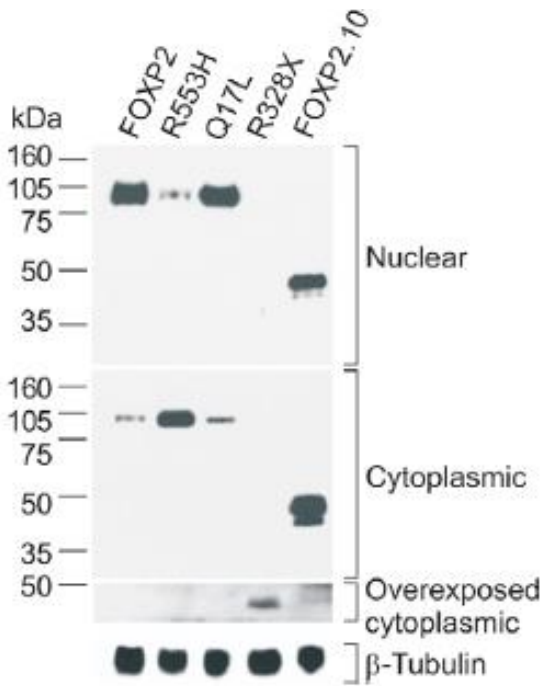
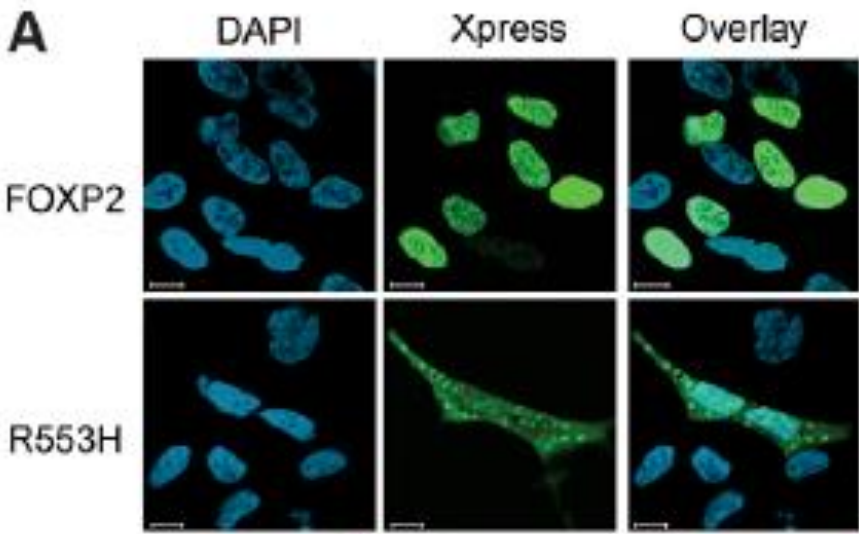


Nucleo-cytoplasmic transport defects during development: FOXP2

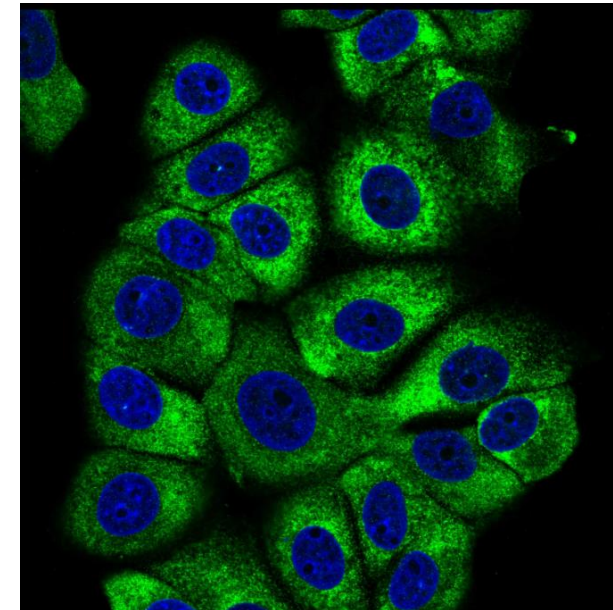
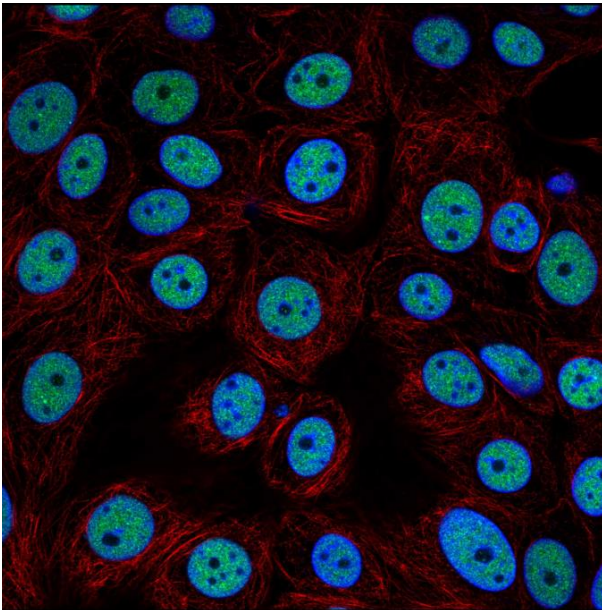
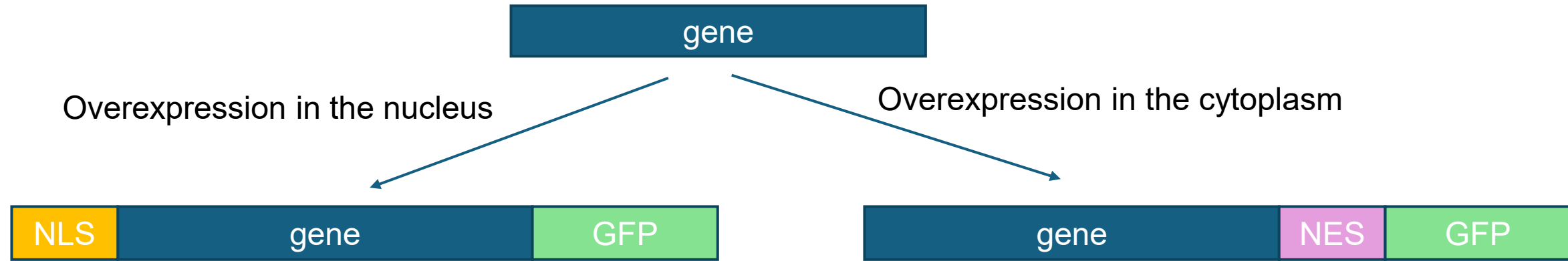


★ NLS

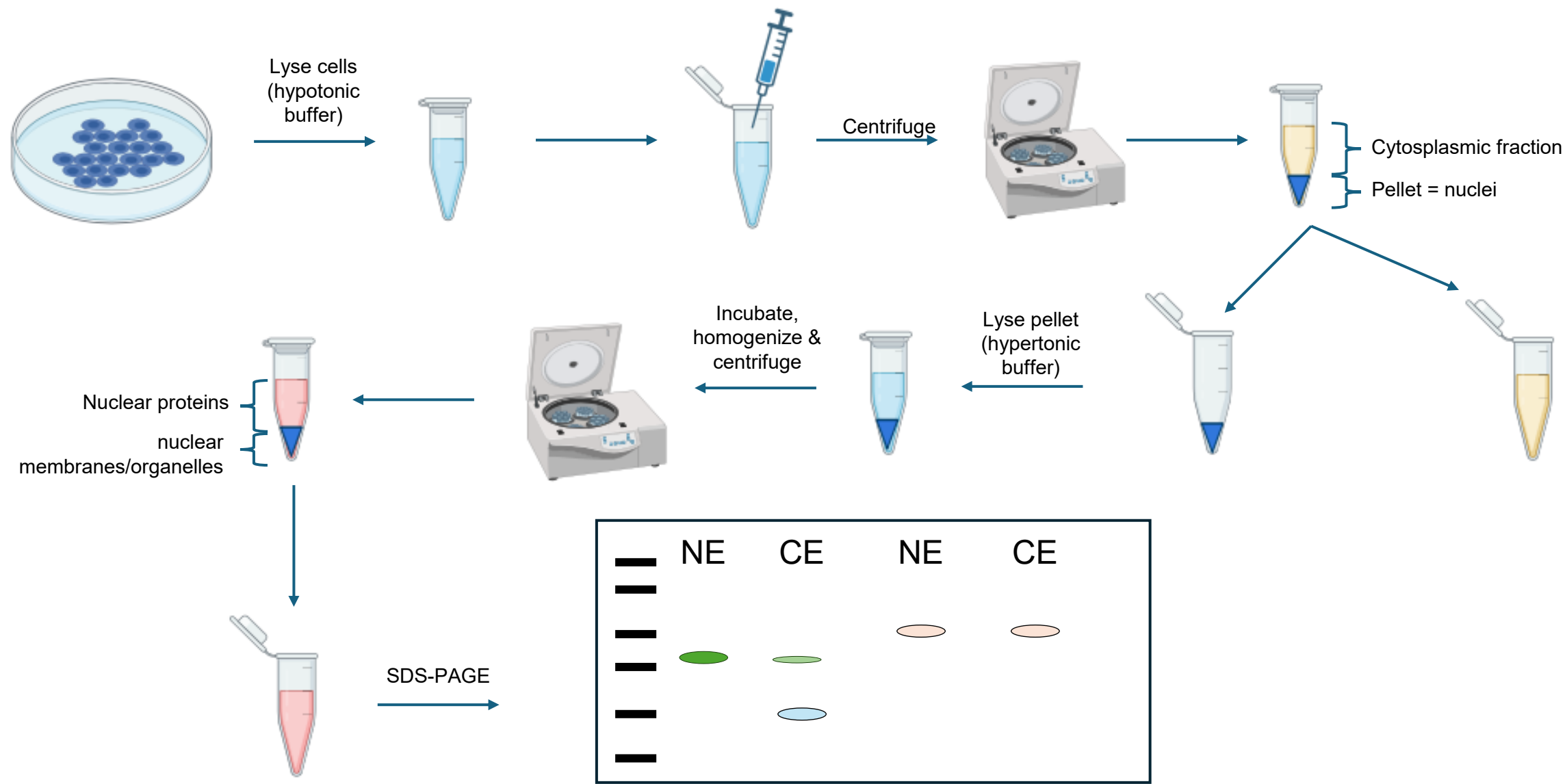
Mutations affecting the NLS of FOXP2 result in impaired learning, disruption of receptive linguistic skills and general cognition.



NLS & NES: practical application



Nuclear/cytoplasmic fractionation



Summary:

- Nuclear Pore Complexes (NPCs) are the gate-keepers at the nuclear/cytoplasmic interface
- Ions and small molecules (<40kDa) can freely diffuse in/out the nucleus.
- Trafficking of RNAs, proteins and large molecules is regulated by the NPCs.
- NLS/NES sequences regulate the protein subcellular localization.
- Neurodegenerative disorders and developmental defects can be caused by defects in the transport system.